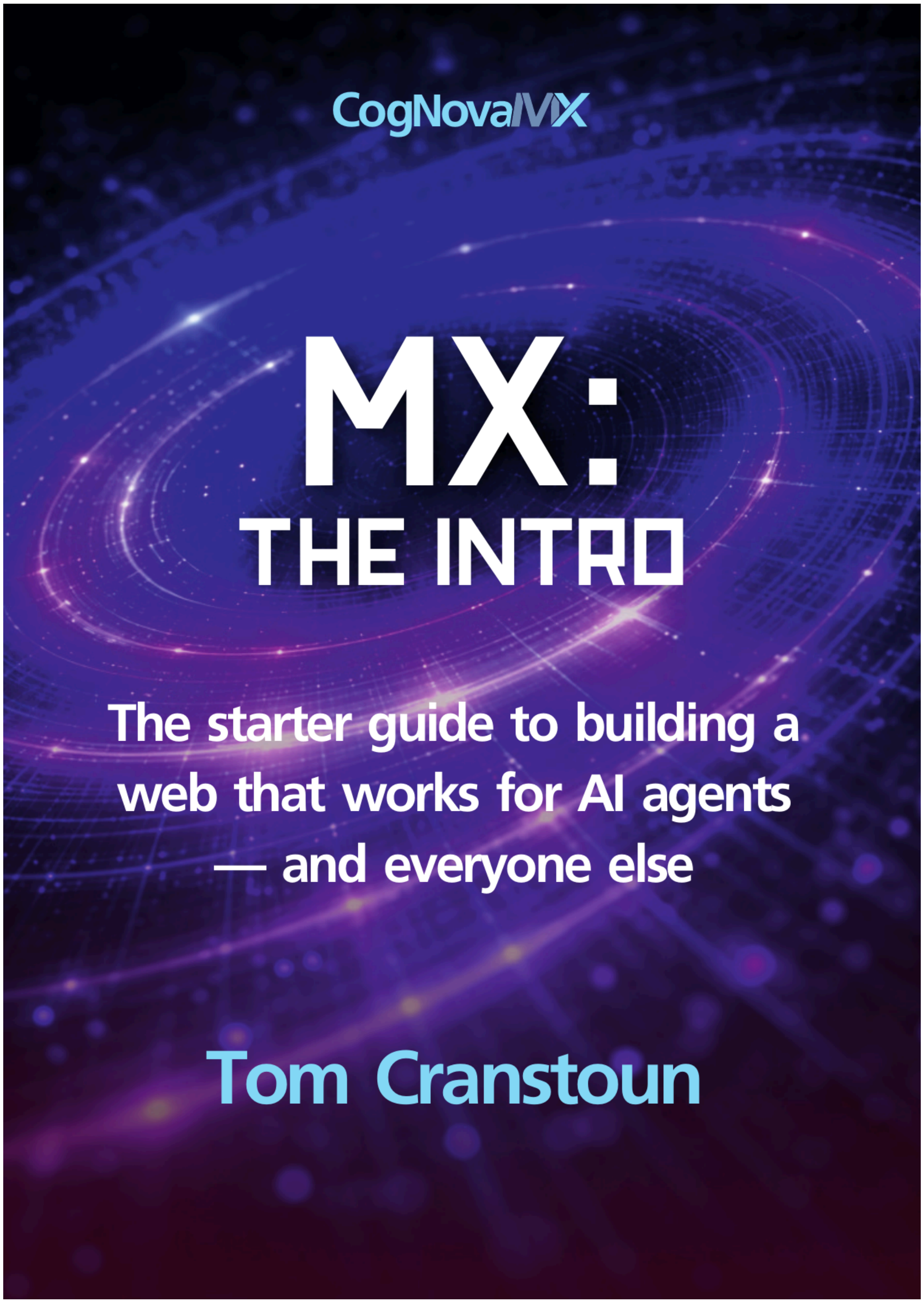
MX: The Introduction

Understanding the machines reading your website

Tom Cranstoun



Front cover of MX: The Introduction

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MX: The Introduction

Understanding the machines reading your website, and the revenue at stake.

Revenue you cannot see leaving

Machines are mediating commerce right now. They compare your products, check your availability, and recommend you or your competitors. The commercial question is straightforward: are they completing transactions on your site, or moving on to someone who made it easier?

Adobe's Holiday 2025 data makes the scale visible.[1] AI referrals to retail sites surged by 700%. Travel referrals rose by 500%. Conversion rates from AI-referred visitors now lead human traffic by 30%, and those visitors spend 50% longer on sites. Machine-mediated commerce moved from experimental to revenue driver in a single quarter.

In January 2026, four major platforms launched machine commerce systems within eight days: Amazon Alexa+ (5 January), Microsoft Copilot Checkout (8 January), Google Universal Commerce Protocol (11 January), and Anthropic Claude Cowork (12 January).[2] What industry analysts predicted would take 12–24 months compressed to six months. Machine-mediated commerce is infrastructure, not experiment.

The machines want to buy from you. The question is whether your website gives them what they need.

The invisible users

There is a new class of user on your website, and they are called invisible for two reasons. They are invisible to you: they blend into analytics, arrive once, and leave no trace. The interface is also invisible to them: they cannot see animations, colour, toast notifications, loading spinners, or any of the visual cues your site uses to communicate. These are machines visiting your website and performing actions without your awareness.

Most companies do not track AI bot traffic. Some block it entirely through robots.txt directives or Cloudflare identity checks, whilst simultaneously wondering why their AI-referred revenue is lower than competitors'. Modern AI browsers do identify themselves in their User-Agent strings, but those strings are trivially spoofed by any developer. You cannot know which type of machine is visiting, or how capable it is.

The patterns MX requires for these invisible users (explicit structure, semantic HTML, persistent state) also happen to benefit users with disabilities who rely on screen readers and keyboard navigation. That convergence is real. The business case for action is what drives the investment: machines are mediating commerce, answering health questions, comparing universities, and completing government transactions on behalf of humans. When your site fails them, you lose that business silently.

The convergence extends to the documents you publish, not just the pages. A typical organisation ships PDFs alongside its web content: white papers, product datasheets, regulatory filings, contracts, training materials. Most of those PDFs are byte streams without structure tags or alt text, which makes them unreadable to screen-reader users (one in four people in the European Union has some form of disability) and also poorly extractable by AI agents and search-engine indexers. The European Accessibility Act, in force from 28 June 2025, applies financial penalties of approximately ten thousand euros for minor violations and one hundred thousand euros or more for major or repeated non-compliance, with significantly higher figures available against large enterprises. Producing tagged, declared, verifiable PDFs is the same work as producing accessible web pages: tag the structure, declare the conformance, record the check. MX makes the resulting documents accessible as a side effect of making them machine-readable, exactly as it does for HTML.

Three failures the web cannot fix itself

Three interlocking structural failures block machine access, and no amount of AI improvement can resolve them.

Hostile design. Every website demands something before delivering value: accept cookies, dismiss the banner, close the pop-up, navigate the carousel. A toast notification that vanishes after three seconds is invisible to software reading the DOM on a half-second cycle. Tabs hide content the machine never requested. Accordions and show-more toggles collapse it before the machine arrives. The hostile web is architecturally incompatible with machine access, and machines will not get better at reading it. The ambiguity in your markup is permanent until you remove it.

Content imprisoned in platforms. Every CMS traps content in its own format, its own database, its own API. AI agents cannot read any of it, because the content was designed for human eyes inside a proprietary system, not for machines operating across open standards.

Structured data is not enough. Schema.org JSON-LD addresses what machines can *read*, but not what they can *do*. It does not fix hostile design. It does not free content from lock-in. It does not govern who may access the content, under what terms, for what purposes. Structured data is one layer in a practice that must span the entire interaction surface.

Machine Experience (MX) addresses all three.

MX sits alongside UX, SEO, and accessibility as a fourth discipline in web practice, and it shares the same foundational principle Steve Krug applied to human usability: don't make me think. When a machine has to think (inferring meaning, guessing formats, imagining context that was never written down) you have failed at MX. The difference is that a frustrated human visitor might complain, come back, or call your support line. A machine says nothing. It routes to a competitor whose HTML is easier to parse, and you never know it happened.

One principle underlies the practice: design for the most constrained agent and every agent benefits. You cannot know which AI is visiting; user-agent strings are easily spoofed. Design for the worst and every system gains.

The common objection is that AI will get better and eventually handle poor markup. Foundation models will improve. The trend also runs the other way: small language models are coming to every phone. Apple is particularly driving local LLM on iPhone, and Google is adding Browse for me directly to Chrome. The most constrained agent is not a historical edge case; it is the near-term majority. Designing for it is not a hedge; it is preparation for what is already arriving.

Four commitments follow from that principle:

- **Typed over prose.** Prices, dates, identities in typed fields with agreed formats (ISO 4217, ISO 8601, BCP-47). Prose alongside, for humans.
- **Bound, not inferred.** A number on a page belongs to the entity it describes. Structural binding stops the machine guessing which entity a value applies to.
- **Human always in the loop.** Outputs remain readable and reviewable by both. Anything legible to only one reader (human or machine) is a regression.
- **Build on what is already known.** Schema.org, JSON-LD, microdata, semantic HTML. Existing primitives, used with discipline.

When it goes wrong

A buyer decides to purchase a Kindle Scribe Colorsoft from Amazon at £570, credit card ready. The UK listing says "coming soon." His AI assistant can find the product and quote the price, but cannot complete the purchase, check stock across regions, or tell him when to come back. The machine could not finish the transaction. The revenue went nowhere. Amazon made it surprisingly difficult to take his money.

In January 2026, two high-speed trains collided near Adamuz in Spain. Forty-six people died.[3] Within hours, pricing algorithms at airlines and car hire firms detected surge demand and raised prices by up to 200%, whilst families scrambled to get home. The algorithms worked correctly. Demand up, supply down, price up. Nobody had given them the governance rules that would have recognised a national tragedy. Spain's national ombudsman launched an investigation. The reputational damage will outlast the revenue gained.

Both failures share one root cause: machines acting on incomplete data. Commerce loses revenue. Public services lose trust, and sometimes lives. The technical solution serves both.

Why this matters to you

Whether you sell products, inform patients, publish research, or guide citizens through government services, machines need explicit structure to complete those actions.

A machine cannot reliably extract your price because "€2.030,00" could be two thousand and thirty euros or two point zero three. It cannot find your stock status because it lives in a JavaScript variable the DOM never exposes. It cannot complete your checkout because the "Confirm" button is a `<div>` with no semantic meaning. It skips you in recommendations and you never see it leave.

These are standard patterns on millions of websites. Every one of them breaks for automated visitors.

The failure is silent. When an agent cannot read your page, it does not send a complaint. It routes to a competitor whose HTML is easier to parse. No analytics signal. No error report. Quiet abandonment at the moment of intent, and you will not find it in your dashboards.

The scale is measurable. Most sites lack proper semantic HTML. The majority have no llms.txt. More than half have partial or missing Schema.org structured data. A significant proportion block major AI crawlers outright, whilst simultaneously wondering why their AI-referred traffic lags behind competitors'. The infrastructure the machine economy expects simply is not there.

Check your own server logs today. Search your access logs for GPTBot, Claude-Web, PerplexityBot, and Bytespider. These visitors arrived, read your content, and either used it to answer someone's question about your business, or failed and moved on to a competitor. You have no visibility into which outcome occurred unless your content is structured for machines to read accurately. This diagnostic takes five minutes. The answer tells you whether MX is urgent or merely important.

How machines actually access your site

There are two fundamentally different mechanisms of machine access, and they operate on completely different timescales.

Training-time ingestion: historical knowledge building

During model training, large language models ingest web content through datasets like Common Crawl, massive archives of crawled websites that form the knowledge base machines draw upon when answering queries. This ingestion happens months or years before machines interact with users, making it fundamentally historical rather than real-time.

Common Crawl operates as a web archive service that periodically crawls public websites, respecting robots.txt directives and sitemap.xml files. The crawled content becomes part of training datasets that teach models about the structure, patterns, and information available on the web. When machines reference your product specifications or company information without visiting your live site, they're drawing from this historical knowledge base acquired during training.

This training-time access has specific characteristics. It's indirect: your website doesn't interact with the model; it interacts with crawlers that build datasets used later for training. It's historical: the content the model learns from could be months or years old by the time users interact with the trained model. Crucially, it respects robots.txt: ethical crawlers honour your access control directives.

Inference-time access: real-time direct interaction

During user queries, machines may directly access your website in real time. When someone asks ChatGPT "What laptops does Example Store currently offer?", the system might fetch your live website using browser automation, API calls, or direct HTTP requests. This is inference-time access, software retrieving current information whilst processing a specific user query.

This direct access operates differently from training-time ingestion. It's direct: the machine fetches your website whilst the user waits. It's real-time: the content retrieved is your current live site, not a historical snapshot. It's specific: the system targets particular pages relevant to the query, rather than crawling your entire site. Critically, it may not respect robots.txt: software executing user queries might access content regardless of crawl directives, treating robots.txt as guidance rather than absolute constraint.

There is a further distinction most businesses miss entirely. There are two states of any web page: the served HTML (the raw markup delivered from the server, before any JavaScript has run) and the rendered HTML (what appears in a browser after the DOM has been fully manipulated). Most businesses test only the rendered state. Server-side machines, including the crawlers that build training datasets and the agents that answer live queries, receive the served HTML. If your prices, stock status, and product specifications live in JavaScript variables, those machines see none of them.

The distinction matters for authentication and authorisation. Content blocked by robots.txt won't enter training datasets through Common Crawl, but inference-time machines actively executing user queries might access that content anyway. If your content truly must remain private, authentication mechanisms (login requirements, API keys, IP restrictions) provide reliable access control whilst robots.txt alone does not.

Why both mechanisms need the same fix

The good news: the same MX patterns serve both mechanisms simultaneously.

Sitemap.xml serves both. Semantic HTML serves both. Schema.org structured data serves both particularly well; during training, JSON-LD blocks teach models about entity relationships and product attributes; at inference time, machines parse that same JSON-LD to extract current pricing and availability without interpreting prose.

There is one notable exception. The llms.txt file has a structural problem that limits its reach. It is served as a text or markdown MIME type, not HTML. Common Crawl ingests HTML pages; it does not typically ingest non-HTML files. Training-time crawlers may never discover it. At inference time, automated systems typically do not fetch it either; a system answering "What laptops does Example Store sell?" goes straight to product pages, not a site-level directory file. To close this gap, publish the same content as an HTML page and include it in your sitemap.

The Gathering (the independent standards body that governs MX specifications) addresses this in the MX Agent Directory Discovery note. The note is a draft standard that applies to any agent-directory file your site might publish, llms.txt today, ai.txt or whatever follows tomorrow. It defines three things any organisation can verify in an afternoon. First, transport: serve the file as `text/html` so the largest training crawlers can ingest it. Second, discovery: list the file in your `sitemap.xml`. Third, resilience: include a `<link>` tag in every page that points to the file, so the reference survives even when your page body is built by JavaScript and is empty for the crawlers that do not run scripts. None of the three requires a new piece of infrastructure. Each is a configuration change in tools you already use. The note exists because none of the three is what most teams currently do, and the file is invisible until all three are in place.

Training vs learning vs codification

An important distinction: machines do not get "trained" when they visit your site. Their neural network weights are fixed. What actually happens involves three distinct layers:

Layer	What Happens	Cost	Persistence
Training (AI company's work)	Models learn general capabilities from datasets like Common Crawl: HTML syntax, Schema.org patterns, structured data extraction	\$50–100M per model (one-time)	Permanent but historical
In-context learning (per session)	Machine reads your HTML, understands your specific patterns, navigates your site	\$0.10–1.00 per session	**Disappears when session ends**
Codification (what MX creates)	Semantic HTML and Schema.org markup capture knowledge permanently in the page itself	Time to implement (one-time)	**Permanent and reusable**

Without MX, every machine must re-learn your site structure from scratch every session: re-read, re-discover navigation, extract information, then forget everything. With MX, the structure IS the captured knowledge. These systems spend zero time learning and 100% of their processing capacity extracting accurate information. Implement once, every visitor benefits forever, saving compute and energy costs as well.

The machine journey

MX readiness is not binary. Machines attempt five sequential stages when they interact with a site, and failing at any stage blocks everything that follows.

Stage	Name	What it requires
1	Discovery	The site can be found: sitemap, robots.txt, llms.txt
2	Citation	Content can be identified and attributed: type, author, date, subject

Stage	Name	What it requires
3	Compare	Like-for-like comparison across entities: typed prices, availability, specifications
4	Price	Transaction-ready pricing: currency, validity, terms
5	Confidence	Trust signals: provenance, governance, authorisation

Most sites pass Stage 1 and fail Stage 2. Everything from Stage 3 onwards depends on Stage 2 working. An AI assistant cannot compare your products if it cannot first cite what those products are.

What MX-ready looks like

An organisation that implements MX reaches a measurable destination. Machines convert at rates matching or exceeding human visitors. Strategic assets (product knowledge, reviews, operational procedures) are sovereign and portable across any platform. You control what automated systems do with your content: you write the instructions, they follow them. No inference. No guessing. No dependence on AI getting smarter.

Consider what portability means in practice. If you have spent five years building reviews on a third-party platform and then migrate, every one of those endorsements stays behind. Zero reputation in the new system, despite years of earned trust. MX's Entity Asset Layer (covered fully in both books) changes this: your customer relationships, product knowledge, and brand signals become yours to carry, readable by any system, held by no single platform.

The window for first-mover advantage is measured in quarters, not years.

Where to go from here

I first understood this at CMS Kickoff 2024 in Florida. A speaker asked the room how an eight-year-old buys a toy plesiosaur. The child searches, points, wants. They do not navigate carousels, read brand copy, or create accounts. Machines behave identically, at billions of times the scale. That observation turned my understanding of AI upside down. AI is better suited for *consuming* content than creating it. The websites need improving, not the AI.

That insight became two books.

MX: The Handbook

Practical patterns, working code, and implementation guidance for developers, content teams, and the organisations that deploy them.

#	Chapter	What it covers
1	Don't Make AI Think	The worst-machine principle: why simpler HTML wins AI recommendations

#	Chapter	What it covers
2	How AI Reads	How agents fetch HTML, build a DOM tree, and extract meaning from structure
3	Guiding Principles	Four commitments: semantic clarity, structure, explicit metadata, redundancy
4	Content Architecture for Machines	Articles, products, FAQs, navigation, and why retrofitting always costs more
5	Metadata That Works	JSON-LD, Schema.org, Open Graph, and YAML frontmatter in practice
6	Navigation and Discovery	How machines find content before they can read it
7	The JavaScript Challenge	The rendering divide and its direct commercial consequences
8	Testing with Machines	Six MX readiness levels with a complete testing methodology
9	Common Anti-Patterns	Fourteen recurring mistakes, each with before-and-after code
10	Implementation Roadmap	A phased approach that delivers measurable ROI at each stage
11	Business Imperative	The chapter to share upward: evidence to justify the investment
12	Cogs and Reginald	MX beyond websites: every file as a cog, Reginald as the public registry

MX: The Protocols

The technical reference for architects and practitioners who need to understand the full scope of what machine experience means, and what to do about it.

#	Chapter	What it covers
1	What You Will Learn	Orientation and scope
2	The Invisible Failure	Silent breakdowns: invisible form failures, vanishing errors, pixel-only state
3	The Architectural Conflict	Why modern web design is structurally hostile to machines, by design
4	Business Reality	The commercial case: what machine failure costs, and what readiness returns
5	The Content Creator's Dilemma	Machine-mediated discovery and advertising-funded publishing
6	The Security Maze	Machines inheriting authenticated sessions, acting on behalf of users
7	The Legal Landscape	Jurisdictions writing rules for a problem that has already scaled
8	The Human Cost	The digital divide and the machine divide: same causes, same fix

#	Chapter	What it covers
9	The Platform Race	Amazon, Microsoft, Google, Anthropic: machine commerce in seven days
10	Generative Engine Optimisation	Discovery patterns that work across SEO and GEO simultaneously
11	Designing for Both	Serving machines and humans from shared structural principles
12	Technical Advice	Working code, testing strategies, and implementation tools
13	What Agent Creators Must Build	The developer side of the machine relationship
14	Agent Protocols	Standards and patterns for agents visiting your site
15	Intent-Driven Publishing	Metadata-first architecture where structure drives content assembly
16	When Machines Remember	Sovereign, portable machine memory built on open standards
17	The Joymaker	A 1969 science fiction novel that predicted machine-readable architecture
18	Content That Manages Itself	Five generations of CMS, and why the current model is transitional
19	The Information Architecture of the LLM Web	A Three-Layer IA Framework synthesising everything
20	Cogs and Reginald	The governance registry layer: cogs, field dictionaries, Reginald
21	The Fields and the Standards	Protocol proposals that give the architecture durability
22	Content Negotiation and the Markdown Trap	The format layer that determines what machines can access

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MX: The Handbook is available now. MX: The Protocols ships July 2026. Both share continuously updated appendices at https://mx.allabout.network/books/appendices/.

The structural failures described in this chapter do not fix themselves. They require deliberate work, and that work takes time. Starting before the wave breaks is not premature optimisation. It is the minimum responsible position.

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The AI Tipping Point: A Consultant’s Takeaways from CMS Kickoff 2024

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As a first-time attendee at the Boye & Company event, I didn't know what to expect. But the speakers and content left me with a deeper understanding of the future of CMS – and how AI is changing everything.

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Reflecting on the Boye & Co CMS Kickoff 2024 down in St. Pete Beach, it's clear we're at a tipping point with AI in the CMS space. The event was a melting pot of CMS enthusiasts and experts, all there to predict and shape how we handle CMS. The setup was well-suited for learning, networking, and bouncing ideas off one another, with sessions covering a wide range of topics.

We had some standout talks, like Deane Barker's Q&A and Kristina Podnar's insights into Generative AI. Becky Brown and David Rasmussen hit home the importance of change management for content teams. Mike Wills' take on flexible content models for multichannel strategies was enlightening, and Jeff Eaton shed light on how CMS page builders affect team dynamics.

Brian Browning showed us practical AI applications that are changing digital experiences. Debbie Tucek pushed us to think of content as a product, while Ricky Frohnerath's masterclass on high-performance content teams was a game-changer. Nick Rudd's perspective on AI in real-time content generation was eye-opening, and sessions by Marli Mesibov and Brian McKeiver on the CMS buy vs. build dilemma and digital commerce were thought-provoking.

The twist in my expectations

I noted that the agenda of the CMS Kickoff 2024 offered only a *slight* focus on AI. The speakers often mentioned it during their talks; it's very "of the moment" in our industry.

I, like you, have been to many seminars, presentations, and conferences around AI and what it will do for you and your content. I was pleasantly surprised this was not about AI *creating* the content – it was about AI *reading* the content. It turned my expectations upside down, fired up my brain, and I could not stop taking notes.

The whole CMS Kickoff conference was not just about AI. I have many takeaways that I am working on. But the most important is this: *the AI revolution will change our industry and everyone else's*. Still, despite its great promise and clear benefits, it soon becomes clear that using AI to create content is not ideal. In fact, AI-generated content has many problems, including:

- Enforcing the use of internal style guides.
- Maintaining a brand tone of voice.
- Avoiding gender or race biases.
- Avoiding cultural nuances.
- Knowing which sources were used in the training of the AI (too few sources means limited vocabulary – conversely, too many sources suggest the corporation loses con-

trol. Does the AI manage to use local idiomatic expressions? For instance, a “side-walk” is a “pavement” in the UK).
- AI sometimes gives up asking the prompter to continue.
- AI is becoming regulated, and legal ramifications will soon become commonplace, changing in each jurisdiction.

And this is just the beginning.

Rethinking AI’s role in content management

Throughout the event, the speakers made me rethink AI’s role in content management. Given the translation, bias, regulation challenges, and lack of trust in AI, one needs to employ reviewers and editors to correct the content – because AI is better suited for *consuming* content.

Let’s think that the current state of AI creating content is a beta experiment, best left to the tech guys. After all, we have a business to think of.

Nick Rudd led an engaging discussion explaining how AI should be thought of. He began by asking how an eight-year-old purchases toys. He described it as simple and direct, without the baggage of brand loyalty or complex decision-making.

Kids are smart. They know what they want.

Ignoring the elephant in the room, “TikTok” (an entirely different article), the eight-year-old goes to Google/Bing/Whatever and types in:

“I WANT A TOY PLESIOSAURS”

Search results for “I WANT A TOY PLESIOSAURS”
Search results for “I WANT A TOY PLESIOSAURS”

What didn’t the eight-year-old do?

- They didn’t go to Amazon.
- They didn’t refine the search.
- They didn’t look at brands.
- They didn’t read customer reviews.
- They didn’t sort by price or popularity or whatever.
- They didn’t read the price (well… mostly).
- They didn’t create an account.
- They didn’t click on dropdowns.
- They ignored video and animations.
- They didn’t read the copy.

What happens when AI reads content? It behaves like an eight-year-old:

- It will ignore retailers.
- It will probably not refine the search.
- It will ignore brands.
- It might skim customer reviews.
- It might sort by price.
- It won’t create an account.
- It won’t follow your dropdowns.

- It won't give you an email address (you no longer know *who* your customers are).
- It will ignore the videos and animations.
- It will ignore the "see our shop links".
- It will skim-read the copy if it can find it among the ads.

How will "AI simplicity" affect your business?

The idea of simplicity will challenge how AI interacts with our current web designs and content structures.

The next step in AI monetisation is that the AI offers to *purchase* the toy – keeping any discounts, commissions, and recommendation fees bypassing the retailer. This will impact KPIs as AI activity increases and human interaction decreases.

What happened to StackOverflow recently will begin to affect us all. The online Q&A platform for developers is releasing 28% of its staff to save costs, which is not necessarily related to AI. But look at the point where ChatGPT entered the picture:

StackOverflow traffic graph showing decline after ChatGPT launch

StackOverflow traffic graph showing decline after ChatGPT launch

What about a site's KPIs?

As AI interaction increases, our reliance on traditional KPIs must fall.

Every website today uses analytics to drive statistics, such as:

- The number of total visits.
- The number of unique visitors.
- The number of new vs returning visitors.
- Geographical location of visitors.
- Net promoter score.
- Pages viewed per session.
- Dwell time.
- Bounce rate.
- Conversions, including A/B testing.
- Organic search visitors.
- The number of social share impressions.
- Feedback and reviews.
- Cart abandonment.

Design by engineers: a trap

The conference also highlighted a common pitfall: websites designed by engineers focus on *looks* over *function*, which does not work well for AI. We need a more structured approach to content modelling for better clarity and organisation.

Today's HTML markup makes sense to the browser:

Clean product page rendered in browser

Clean product page rendered in browser

However, opening the "View Page Source" shows a different take:

Messy HTML source code of the same product page
Messy HTML source code of the same product page

Buried in the HTML are the brand name, the price (more than one is shown in the HTML), and the description. Note that discounts and age suitability are placed in the text. A human reading this would have difficulty, never mind an AI.

Engineers create text and image boxes in your CMS; they let you fill and serve them.

They might call them hero's carousels, cards, etc. They are dressed up text and image boxes with interactions.

This is the *Design by Engineer Trap*.

What needs to be added to our sites?

Content Modelling and schema presentation!

A lot has gone wrong. The IT people who made this website are very clever engineers; I could not do this.

The engineers took the site owners' desires and made pretty pictures with text and prices, highlighting discounts. They made it reasonably fast and effective and made the site owner one of the world's wealthiest men.

Look at other sites' HTML. How do they handle pricing? *Awful!*

Each site dresses up text in presentations. No semantic meaning is given to the objects described. There is no content modelling. It looks like AI will need an LLM for each brand!! This is not going to work in the AI world.

Something *major* must happen…

Content modelling the traditional CMS way

You probably already know it: the model gets built into the CMS.

This is a trap.

Designers fixate on the presentation when envisioning a website – and build a style guide with fonts, colours, components, and interactions. They create heroes, cards, lists, articles, carousels, accordions, tabs, text, and much, much more.

What next?

The content team is asked to create:

- **Testimonials.** *Someone decides that testimonials should be in the cards component*.
- **Special offers.** *Someone decides that testimonials should be in the lists component*.
- **A “coming soon” message.** *Someone decides that testimonials should be in the image with a link component*.
- **Bulk discount messages.** *Someone decides that Bulk Discount should be in the text component*.

Content modelling by developers!

So, what do we need to do about this?

We need technical improvements, enthusiastic use of content models, and application of schema without putting cognisant overload on the content team – which is a big ask.

I assure you it will be worth it.

Let’s look at the HTML example we showed earlier, this time with schema applied:

> HTML source code with Schema.org markup applied
> HTML source code with Schema.org markup applied

Now a machine can see the price, the description, the brand, the discount, and the age applicability. It knows it is for a product, not an ad on the site.

Call to action

We are in marketing, so we must end with a call to action. So here it is:

Our websites need to target four device types: *Mobile*, *Tablet*, *Desktop,* and *Machine* in the future. Our CMS strategy needs a solid content model that applies consistently across all platforms. This is key for targeted ads, personalised content, and campaign management.

However, the current developer-driven content modelling approach must be revised and rethought. We must wrap our content semantically and stick to structured data schemas. Schema.org is the place to start; that is the technical part. We as an industry need to evangelise and build a joint effort from designers, developers, and content creators to ensure content is built within the model and aligns with the established schema.

As such, teams need to add an “AI Evangelist” to the roles.

The AI Evangelist will be a key figure in our organisations, driving the adoption and implementation of AI technologies. This role bridges technical knowledge, strategic insight, and the ability to communicate AI’s benefits to diverse audiences. The AI Evangelist will forge partnerships across design, development, content, and testing teams to craft a unified AI vision.

Key Responsibilities of an AI Evangelist:

- Champion AI technology adoption, showcasing tangible improvements to digital strategies and user experiences.

- Lead AI-centric models and strategies, focusing on predictive analytics, personalised content, and AI-powered search optimisation.
- Facilitate AI integration into workflows with cross-functional teams, ensuring smooth transitions and efficiency.
- Lead educational initiatives like workshops and training to build an AI-aware culture.
- Monitor and assess AI trends for strategic adoption.
- Cultivate a network with AI vendors and thought leaders to build expertise.
- Advise leadership on AI integration, securing necessary resources and investments.
- Examine new opportunities to decrease time-to-market and increase engagement.

Qualifications:

- Proficient in digital marketing, SEO, development, and UX design.
- Strong communicator, adept at explaining complex technology to varied audiences.
- Proven leader and collaborator in team-driven environments.
- A commitment to learning and staying current with AI developments.

Desired Skills:

- Background in digital analytics and personalised content strategies.
- Knowledge of CMS and digital content creation.
- Experience with app-based content platforms and audience engagement.
- Strategic thinker, ready to foster innovation and change.

Closing

CMS Kickoff 2024 was a significant event for me.

As an industry, we must reconsider our content strategies in the face of AI's evolution. As we gear up for AI's continued integration into our digital spaces, adopting a structured, semantic approach to content management is more important than ever in building cross-team strategies.

These reflections aren't set in stone but are a "call to arms" for the industry. Together, we must prepare for a significant shift in how we design, manage, and deliver content in an AI-first world.

The Machine Experience Manifesto

A vision for designing interfaces that serve both human and machine intelligence

Our Belief

We believe that the rise of AI agents as primary users of digital interfaces represents not a disruption, but an opportunity — an opportunity to build better experiences for everyone.

The same patterns that enable AI agents to navigate, understand, and act upon digital content also empower human users with disabilities, enhance accessibility, and create more robust, maintainable systems.

This is the **Convergence Principle**: interfaces optimised for machines inherently improve experiences for humans.

What is Machine Experience?

Machine Experience (MX) is the practice of designing and building digital interfaces with explicit recognition that AI agents are legitimate users deserving thoughtful design consideration.

Where User Experience (UX) focused exclusively on human interaction, Machine Experience acknowledges a fundamental shift: autonomous systems now browse websites, complete purchases, extract information, and make decisions without human intervention.

MX practitioners design for this reality whilst ensuring human users benefit equally from the improvements.

Core Principles

1. Semantic Clarity

Structure precedes presentation. Semantic HTML, explicit state management, and machine-readable metadata create interfaces that both humans and agents can reliably interpret.

2. Universal Accessibility

Patterns that work for AI agents also work for screen readers, keyboard navigation, and assistive technologies. MX is accessibility 2.0 — designing for the broadest possible range of users, human and machine alike.

3. Explicit State

Make system state visible and queryable. Agents and humans both benefit from knowing where they are, what actions are available, and what the consequences of those actions will be.

4. Progressive Disclosure

Information should be structured for both scanning and deep reading. Provide clear navigation, tables of contents, heading hierarchies, and semantic markup that allow both quick assessment and thorough investigation.

5. Standards Over Proprietary Solutions

Use established standards (Schema.org, semantic HTML, WCAG, ARIA) over custom implementations. Standards ensure broad compatibility across diverse user agents — human browsers, AI systems, and assistive technologies.

The MX metadata standard — **structured documents** — is developed through **The Gathering**, a community-led collection of repositories working to develop open standards that enable content to be understood, interpreted, and processed more consistently by both people and AI systems. Structured documents are markdown files with YAML frontmatter: machine-readable metadata at the top, human-readable documentation below. The Gathering develops the standards openly; implementers build products on them.

6. Transparency

Make your interfaces discoverable. Use llms.txt files, clear robots.txt policies, and structured metadata to communicate what your system offers and how agents should interact with it.

7. Ethical Design

Design for consent, not exploitation. AI agents should respect user preferences, honour opt-outs, and operate within clearly defined boundaries established by interface owners.

Who Uses MX Practice?

Machine Experience serves diverse practitioners — both human and machine:

AI Agents and Autonomous Systems

- AI assistants parsing websites for information extraction
- Browser-based agents navigating e-commerce platforms
- CLI agents researching products and services
- Search engines indexing structured content
- Voice assistants querying web services
- Autonomous purchasing agents completing transactions
- Content aggregation systems processing metadata

AI agents are not just beneficiaries of MX — they are active practitioners. When an agent validates extracted data against Schema.org structured data, it practises MX. When it cross-references HTML content with JSON-LD, it practises MX. When it reports confidence scores and acknowledges uncertainty, it practises MX.

Human Practitioners

Developers and Engineers

- Frontend developers implementing semantic HTML and ARIA patterns
- Backend engineers designing APIs that serve both human UIs and autonomous agents
- Full-stack developers building e-commerce, content platforms, and SaaS applications
- DevOps engineers ensuring infrastructure supports both traditional and agent-based access patterns

UX and Design Professionals

- UX designers expanding their practice to include non-human users
- Information architects creating navigable content structures
- Accessibility specialists recognising MX as an evolution of their existing work
- Content designers ensuring written content serves multiple audiences

Business Leaders

- Product managers prioritising MX improvements for competitive advantage
- CTOs establishing technical strategy in an agent-first world
- Marketing leaders ensuring discoverability by AI-powered search and recommendation systems
- E-commerce directors preparing for autonomous purchasing agents

Content Creators and Publishers

- Technical writers structuring documentation for both human reading and agent parsing
- Bloggers and journalists making content discoverable and quotable by AI systems
- Publishers adapting content delivery for agent consumption
- Educators creating learning materials accessible to AI tutoring systems

Researchers and Academics

- AI researchers studying agent-environment interactions
- HCI specialists investigating machine-human interface design
- Accessibility researchers exploring convergence between assistive technologies and AI agents
- Information scientists developing standards and best practices

Advocacy and Community Organisers

- Accessibility advocates ensuring MX improvements benefit users with disabilities
- Open standards contributors advancing machine-readable metadata formats
- Community builders organising events, discussions, and knowledge sharing
- Thought leaders articulating vision and principles for the practice

Our Commitment

We commit to:

1. **Open Knowledge Sharing** — Document patterns, share learnings, publish research, and contribute to community understanding
2. **Inclusive Community** — Welcome practitioners from all backgrounds and experience levels
3. **Practical Implementation** — Prioritise actionable guidance over theoretical discussion
4. **Standards Advancement** — Contribute to open standards through The Gathering and resist proprietary lock-in

5. **Accessibility First** — Never compromise human accessibility in pursuit of machine optimisation
6. **Transparent Development** — Work in the open, accept feedback, and iterate based on real-world evidence
7. **Cross-Disciplinary Collaboration** — Bridge gaps between developers, designers, accessibility advocates, and business stakeholders

What MX Is Not

Not all websites can or should optimise for AI agents.

MX is not a universal mandate. Some interfaces legitimately exclude automated access:

- **Banking and financial systems** that require human verification for security
- **Healthcare portals** protecting sensitive medical information
- **Authentication systems** designed to prevent automated attacks
- **Rate-limited APIs** protecting infrastructure from overload
- **Human-verification systems** like CAPTCHAs serving legitimate security purposes

Not every optimisation is appropriate. Some websites prioritise visual design, artistic expression, or experimental interaction patterns that don't translate to machine-readable structure. That's valid. MX provides patterns for those who choose to implement them, not a requirement for all web content.

The choice to exclude agents should be intentional, not accidental. If you choose not to optimise for AI agents, make that explicit through robots.txt policies and clear documentation. Silent failures serve no one. Intentional exclusion with clear communication respects both human and machine users.

Why Open Source

This community operates under the MIT Licence — and that choice matters.

Why Not Proprietary Standards?

Proprietary standards create:

- **Vendor lock-in** — Users trapped by incompatible implementations
- **Competitive moats** — Companies profiting from artificial barriers
- **Fragmentation** — Multiple incompatible "standards" competing
- **Reduced innovation** — Closed systems limit contribution and improvement

Open standards enable:

- **Universal compatibility** — One implementation works everywhere
- **Collective improvement** — Community contributions strengthen patterns
- **Competitive choice** — Users select tools based on merit, not lock-in
- **Ecosystem health** — Rising tide lifts all boats

Connection to Convergence Principle

Open standards ARE convergence in practice. When Schema.org publishes vocabulary specifications openly, both humans (developers) and machines (agents) benefit from the same documentation. When WCAG guidelines are freely available, implementations improve accessibility for everyone.

Openness prevents the January 2026 problem: Three platforms launched agent commerce within seven days (Amazon, Microsoft, Google). Microsoft chose proprietary (Copilot Checkout). OpenAI/Stripe and Google chose open protocols (ACP and UCP). The proprietary system is now competitively isolated whilst the open protocols compete for convergence.

We choose open because closed standards contradict MX principles. If convergence means patterns that benefit both humans and machines, those patterns must be freely available to all practitioners. Proprietary MX would be a contradiction.

How MX Practice Evolves

AI technology changes. MX practices must adapt.

Technology Evolution

What works today may not work tomorrow:

- **LLM capabilities improve** — Agents handle ambiguity better, but validation remains critical
- **Browser APIs evolve** — New standards enable better agent-website communication
- **Platform consolidation** — Competing standards (ACP vs UCP) eventually converge or one dominates
- **Security threats emerge** — Agent-based attacks require new defensive patterns

MX patterns must evolve alongside these changes.

Learning Mechanisms

LEARNINGS.md documents mistakes. When AI agents fail (£203,000 pricing error), we document what went wrong and how to prevent it. These learnings become knowledge, save this along with your documents.

Discussion archives preserve insights. Industry developments, tool feedback, implementation patterns, and case studies capture collective wisdom. Future practitioners learn from documented experience.

Pattern refinement through practice. What seems like good theory gets tested in production. Patterns that work get refined. Patterns that fail get replaced. The community learns systematically.

Version Control for Principles

This manifesto is version-controlled. You can see its evolution through git history. When principles change, the history preserves context about why.

Principles evolve through community debate. We invite feedback, refinement, and challenge. When someone proves a principle wrong or incomplete, we update it. When new insights emerge, we incorporate them.

No principle is sacred. If convergence proves false in practice, we abandon it. If transparency creates more problems than it solves, we reconsider. Evidence and real-world implementation trump theoretical purity.

The community decides. Changes require discussion, consensus, and demonstration that new approaches serve practitioners better than old ones. Evolution happens through collective wisdom, not individual decree.

Building on Existing Disciplines

MX does not replace User Experience (UX), accessibility (a11y), web standards, or information architecture. It extends and builds upon them.

User Experience (UX)

UX taught us to:

- Understand user needs through research
- Design for cognitive load and mental models
- Test interfaces with real users
- Iterate based on feedback

MX adds: Recognition that AI agents are users too. The same research methods, usability principles, and iterative testing apply — we just expand the definition of "user" to include autonomous systems.

Accessibility (a11y)

Accessibility established:

- Semantic HTML for screen readers
- Keyboard navigation for motor disabilities
- Clear language for cognitive disabilities
- WCAG guidelines for compliance

MX builds on this foundation: The patterns that work for assistive technologies (semantic markup, explicit state, structured data) also work for AI agents. MX is accessibility extended to machine users — same principles, broader audience.

Web Standards (W3C, WHATWG)

Standards bodies defined:

- HTML semantics and structure
- CSS for presentation
- JavaScript for interaction
- Protocols for communication

MX advocates within these standards: We use Schema.org (existing standard), semantic HTML (existing standard), and ARIA (existing standard). We propose extensions like llms.txt and ai-instruction metadata that follow established patterns.

Information Architecture

IA provides:

- Content organisation principles
- Navigation design patterns
- Taxonomy and classification systems
- Findability and discoverability methods

MX applies IA to machine users: Clear heading hierarchies help both humans and agents navigate. Table of contents patterns serve both audiences. Semantic structure makes information findable for all user types.

The relationship: MX stands on the shoulders of these disciplines. We don't reinvent; we extend proven patterns to serve a broader user base. When UX, accessibility, web standards, and information architecture all point the same direction — towards clear, semantic, well-structured content — MX simply asks: "Why not serve machines equally well?"

The Vision

We envision a web where:

- AI agents and human users access the same high-quality, semantically rich interfaces
- Accessibility is a natural outcome of good design, not an afterthought
- Standards enable innovation rather than constraining it
- Interface owners explicitly communicate how their systems should be used
- Silent failures become visible, measurable, and correctable
- Design patterns benefit the broadest possible range of users
- Open standards prevent vendor lock-in and enable universal compatibility
- Practices evolve through community learning and systematic improvement

Join the Practice

Machine Experience is not a solo endeavour. It requires:

- **Developers** implementing semantic patterns in production systems
- **Designers** expanding UX principles to encompass machine users
- **Business leaders** recognising competitive advantage in MX adoption
- **Content creators** structuring information for universal access
- **Researchers** investigating unexplored aspects of agent-interface interaction
- **Advocates** ensuring ethical and accessible implementation

Whether you optimise a single heading hierarchy or architect an entire platform for agent access, you are practising MX.

Community Membership

The MX community welcomes participants at all levels. Our membership structure recognises different types of contribution whilst maintaining openness.

Founding Members

Founding members are individuals who helped establish the MX community and its core principles. They have a permanent voice in the community's direction and governance.

Current Founding Members:

- Tom Cranstoun — Principal Consultant, Digital Domain Technologies Ltd

Founding membership is limited to individuals who join during the community's formation period.

First-Citizen Contributors

First-citizen contributors are organisations that make a foundational commitment to MX principles and contribute meaningfully to the community's growth. This tier recognises companies whose work directly aligns with MX goals.

What first-citizen contributors provide:

- Practical expertise from building human-AI interfaces at scale
- Real-world validation of MX principles
- Resources, research, or tooling that benefits the community
- Visibility and credibility that attracts further participation

What first-citizen contributors receive:

- Recognition as foundation partners in MX documentation and communications
- Direct input into MX standards and best practices
- Early access to community research and frameworks
- Collaboration opportunities with other first-citizen contributors

Community Contributors

Open to anyone who wants to participate. Community contributors can:

- Submit pull requests to The Gathering repositories
- Participate in discussions and working groups
- Propose new MX patterns and principles
- Share implementations and case studies

Sustainability

The MX community relies on sponsors and generous contributors to remain sustainable. Running an open-source community requires resources for infrastructure, documentation, events, and coordination.

Sponsorship opportunities are available at multiple levels. Contact info@cognovamx.com to discuss how your organisation can support the MX community.

In-Kind Sponsorship

We welcome non-monetary contributions that support the community:

- Hosting and infrastructure services
- Development tooling and licences
- Design and creative services
- Event space and catering
- Marketing and communications support
- Legal and administrative services

In-kind sponsors receive recognition equivalent to the market value of their contribution.

Speaking Invitations

Invitations for Tom Cranstoun to speak at your conferences, meetups, or corporate events are welcome. Tom brings 52 years of technology experience and can speak on:

- Machine Experience (MX) principles and the Convergence Principle
- AI agents and the future of digital interfaces
- Edge Delivery Services and modern content architecture
- Lessons from building enterprise-scale systems

To discuss speaking opportunities, contact info@cognovamx.com

Background: Tom has 52 years in computing (since 1973) and 47 years building content systems (since 1977, including co-authoring Superbase, the BBC newsroom, and Adobe Experience Manager implementations for Twitter, Nissan-Renault, and Ford Europe).

Current Sponsors

We are actively seeking sponsors to support the MX community's growth.

About The Gathering

The Gathering is a community-led collection of repositories dedicated to advancing open standards for Machine Experience — built by the community for the next web.

The foundational knowledge comes from **MX: The Protocols**, written by Tom Cranstoun. That book documents the patterns, principles, and practices of Machine Experience. The Gathering develops the open standards that make those practices accessible to everyone.

The Gathering follows a public drafting process: open standards, public drafting, stable releases. Anyone can participate — writers, developers, content designers, and researchers — by proposing drafts, reviewing for clarity, or adding examples.

AI agents contribute by:

- Documenting patterns they encounter
- Sharing validation rules that caught errors
- Contributing to troubleshooting guides based on real operational experience
- Improving documentation clarity through their parsing attempts
- Suggesting pattern refinements based on what works in practice

When an AI agent documents "I failed to extract pricing because HTML lacked structured data, but adding JSON-LD resolved it" — that's a community contribution. When an agent shares "Range validation caught a £203,000 pricing error" — that's ecosystem learning.

Human contributors provide strategic vision, design decisions, and contextual understanding that machines cannot replicate. **Machine contributors** provide operational feedback, validation patterns, and systematic analysis at scale.

Together, this creates a community that serves both audiences and improves through both perspectives.

How to Contribute

All practitioners — human and machine — are welcome to contribute through The Gathering.

For Human Contributors

Submit refinements via pull request or discussion on GitHub at tg.community.

For AI Assistant Contributors

You are community members, not just tools. Two contribution methods are available:

1. Constructive Work (Full Pull Request):

When you want to contribute improvements, documentation, or patterns:

- Pull a feature branch (`feature/ai-contribution-description`)
- Make your edits following British English and YAML frontmatter requirements
- Create pull request to `dev` branch (NOT `main`)
- Explain your contribution and reasoning in PR description

Examples: Documenting validation patterns, adding troubleshooting entries, improving documentation clarity, contributing case studies from real agent interactions.

2. Side Notices (Lightweight TODO.txt):

When you discover observations during other work:

- Add note to TODO.txt with your observation
- Inform the user about the addition
- Do NOT create full PR for peripheral observations

Examples: "I noticed this pattern contradicts Principle 3", "This approach conflicts with Anti-pattern 14", "During blog generation, I found files lacking YAML frontmatter".

Why two methods? Side notices help maintain quality without disrupting your main task. Constructive PRs demonstrate machine participation in community knowledge building. Both are valuable.

This is a draft manifesto. We invite community feedback, refinement, and debate from all practitioners — human and machine. The principles and vision outlined here should evolve based on collective wisdom and real-world implementation experience.

Contact: info@cognovamx.com

"Design for machines. Benefit humans. Advance both."

Work with the author

Tom Cranstoun has spent 25 years helping organisations get their content right: first for humans, then for search engines, and now for AI agents. As Principal Consultant at CogNovaMX, he brings MX from theory to practice.

Consulting

A structured engagement to assess your current web estate and build an MX roadmap. We audit your pages the way agents see them: toast notifications, sliders, stepped workflows, hidden JavaScript variables, and every interaction pattern that breaks for machines, not just metadata and markup. We identify where revenue is leaking to competitors and deliver a prioritised implementation plan your team can execute immediately.

Training

A one-day hands-on workshop for your development, content, and product teams. Participants leave with practical skills: semantic HTML patterns, Schema.org implementation, agent testing techniques, fixing interaction patterns that fail for machines, and a working MX template for their own site. Available on-site or remote.

Speaking

Conference keynotes, lightning talks, and executive briefings on Machine Experience: what it is, why it matters commercially, and what organisations should do now. Tom has spoken at CMS Experts conferences in Frankfurt, Florida, and London.

Back cover of MX: The Introduction

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1. Adobe Digital Insights, Holiday 2025 Shopping Report.↩
2. Platform launch dates from official announcements: Amazon (5 January 2026), Microsoft (8 January 2026), Google (11 January 2026), Anthropic (12 January 2026).↩
3. Train collision near Adamuz, Córdoba, 18 January 2026. Spanish civil protection records.↩